Course Project

🎥 Projects how-to (watch it!)

Objective

The goal of this project is to apply everything we have learned in this course to build an end-to-end data pipeline.

Problem statement

Develop a dashboard with two tiles by:

- Selecting a dataset of interest (see Datasets)
- Creating a pipeline for processing this dataset and putting it to a datalake
- Creating a pipeline for moving the data from the lake to a data warehouse
- Transforming the data in the data warehouse: prepare it for the dashboard
- Building a dashboard to visualize the data

Data Pipeline

The pipeline could be **stream** or **batch**: this is the first thing you'll need to decide

- **Stream**: If you want to consume data in real-time and put them to data lake
- **Batch**: If you want to run things periodically (e.g. hourly/daily)

Technologies

You don't have to limit yourself to technologies covered in the course. You can use alternatives as well:

- **Cloud**: AWS, GCP, Azure, ...
- **Infrastructure as code (IaC)**: Terraform, Pulumi, Cloud Formation, ...
- **Workflow orchestration**: Airflow, Prefect, Luigi, ...
- **Data Warehouse**: BigQuery, Snowflake, Redshift, ...
- **Batch processing**: Spark, Flink, AWS Batch, ...
- **Stream processing**: Kafka, Pulsar, Kinesis, ...

If you use a tool that wasn't covered in the course, be sure to explain what that tool does.

If you're not certain about some tools, ask in Slack.

Dashboard

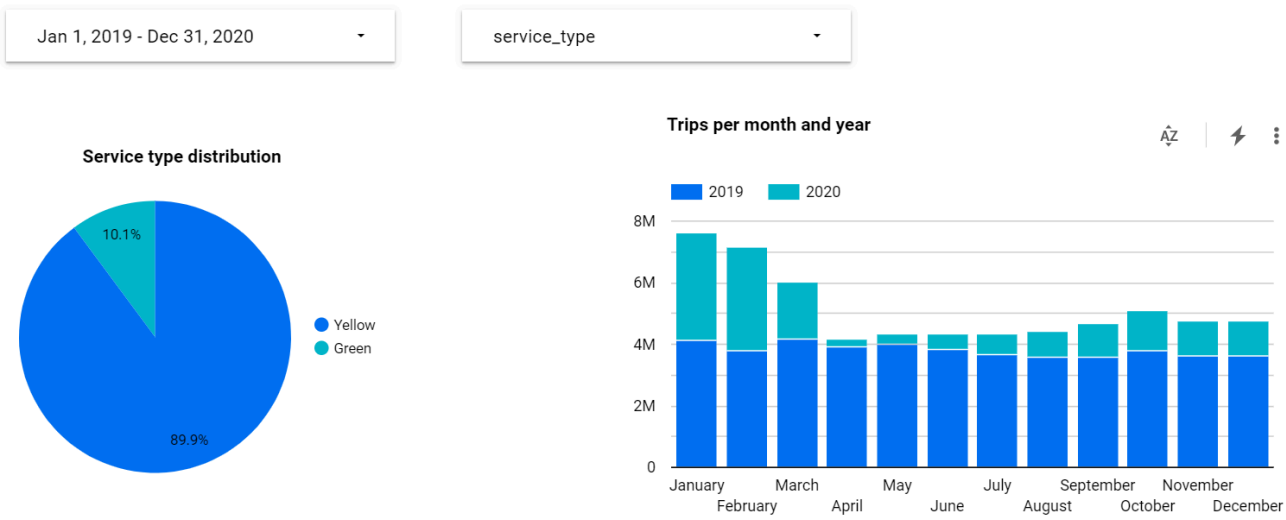
You can use any of the tools shown in the course (Looker Studio or Streamlit) or any other BI tool of your choice to build a dashboard. If you do use another tool, please specify and make sure that the dashboard is somehow accessible to your peers.

Your dashboard should contain at least two tiles, we suggest you include:

- 1 graph that shows the distribution of some categorical data
- 1 graph that shows the distribution of the data across a temporal line

Ensure that your graph is easy to understand by adding references and titles.

Example dashboard:



Peer reviewing

> **Important**
>
> To evaluate the projects, we'll use peer reviewing. This is a great opportunity for you to learn from each other.
>
> - To get points for your project, you need to evaluate 3 projects of your peers

> - You get 3 extra points for each evaluation

Evaluation Criteria

- Problem description
  - 0 points: Problem is not described
  - 2 points: Problem is described but shortly or not clearly
  - 4 points: Problem is well described and it's clear what the problem the project solves
- Cloud
  - 0 points: Cloud is not used, things run only locally
  - 2 points: The project is developed in the cloud
  - 4 points: The project is developed in the cloud and IaC tools are used
- Data ingestion (choose either batch or stream)
  - Batch / Workflow orchestration
    - 0 points: No workflow orchestration
    - 2 points: Partial workflow orchestration: some steps are orchestrated, some run manually
    - 4 points: End-to-end pipeline: multiple steps in the DAG, uploading data to data lake
  - Stream
    - 0 points: No streaming system (like Kafka, Pulsar, etc)
    - 2 points: A simple pipeline with one consumer and one producer
    - 4 points: Using consumer/producers and streaming technologies (like Kafka streaming, Spark streaming, Flink, etc)
- Data warehouse
  - 0 points: No DWH is used
  - 2 points: Tables are created in DWH, but not optimized
  - 4 points: Tables are partitioned and clustered in a way that makes sense for the upstream queries (with explanation)
- Transformations (dbt, spark, etc)
  - 0 points: No tranformations
  - 2 points: Simple SQL transformation (no dbt or similar tools)
  - 4 points: Tranformations are defined with dbt, Spark or similar technologies

- Dashboard
  - 0 points: No dashboard
  - 2 points: A dashboard with 1 tile
  - 4 points: A dashboard with 2 tiles
- Reproducibility
  - 0 points: No instructions how to run the code at all
  - 2 points: Some instructions are there, but they are not complete
  - 4 points: Instructions are clear, it's easy to run the code, and the code works

> **Note**
>
> It's highly recommended to create a new repository for your project (not inside an existing repo) with a meaningful title, such as "Quake Analytics Dashboard" or "Bike Data Insights" and include as many details as possible in the README file. ChatGPT can assist you with this. Doing so will not only make it easier to showcase your project for potential job opportunities but also have it featured on the Projects Gallery App. If you leave the README file empty or with minimal details, there may be point deductions as per the Evaluation Criteria.

Going the extra mile (Optional)

> **Note**
>
> The following things are not covered in the course, are entirely optional and they will not be graded.

However, implementing these could significantly enhance the quality of your project:

- Add tests
- Use make
- Add CI/CD pipeline

If you intend to include this project in your portfolio, adding these additional features will definitely help you to stand out from others.